

Chapter 36: Inner Product Spaces Worksheet

1. Are each of the following statements true or false?

- (a) Suppose V is an inner product space, and $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ for every vector $\mathbf{v}$ in V , then $\mathbf{u} = \mathbf{0}$.
- (b) Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$, the function $q : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $q(\mathbf{u}, \mathbf{v}) = u_1v_1 - u_2v_2$ is an inner product in $\mathbb{R}^2$.
- (c) Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$. The function $q : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as $q(\mathbf{u}, \mathbf{v}) = u_1v_2$ is an inner product on $\mathbb{R}^2$.
- (d) The function $q : M_{22} \times M_{22} \rightarrow \mathbb{R}$ defined as $q(A, B) = \text{tr}(AB)$ is an inner product on M_{22} .

2. Consider P_2 , and define

$$\langle f, g \rangle = f(0)g(0) + f(1)g(1) + f(2)g(2)$$

Show that this is an inner product. More specifically, show each property holds:

- (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}$ in V .
- (b) $\langle \mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
- (c) $\langle r\mathbf{u}, \mathbf{v} \rangle = r\langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}$ and $\mathbf{v}$ in V and r in $\mathbb{R}$.
- (d) $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$.

Note: Property (4) is a bit tricky. Think about what you know about 2nd-degree polynomials and their roots.

3. We define an inner product on P_1 by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$.

Consider the following pair of vectors. Compute the inner product of $f(x)$ and $g(x)$, then find the unit vector that is a positive multiple of $f(x)$.

$$f(x) = 1 + x, g(x) = 1 - x$$

4. Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be vectors in $\mathbb{R}^2$, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 5u_2 v_2$$

Show that this is an inner product. More specifically, show each property holds:

- (a) $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ for all $\mathbf{u}, \mathbf{v}$ in V .
- (b) $\langle \mathbf{u} + \mathbf{w}, \mathbf{v} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V .
- (c) $\langle r\mathbf{u}, \mathbf{v} \rangle = r\langle \mathbf{u}, \mathbf{v} \rangle$ for all $\mathbf{u}$ and $\mathbf{v}$ in V and r in $\mathbb{R}$.
- (d) $\langle \mathbf{v}, \mathbf{v} \rangle > 0$ for all $\mathbf{v} \neq \mathbf{0}$.

5. Let A, B be 2×2 matrices, and define $\langle A, B \rangle = \text{tr}(AB^T)$.

Find the distance between $A = \begin{bmatrix} 2 & 0 \\ -3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 1 & -3 \end{bmatrix}$.